Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/27/2006
PI Name:	Hornberger, Troy A. Ph.D.		
Project Title:	Regulation of protein synthesis by strain induced signaling events		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	thornb1@uic.edu	**Fax:**	FY 312-996-2958
PI Organization Type:	UNIVERSITY	**Phone:**	312-996-6353
Organization Name:	University of Illinois at Chicago		
PI Address 1:	School of Kinesiology		
PI Address 2:	901 West Roosevelt Road		
PI Web Page:			
City:	Chicago	**State:**	IL
Zip Code:	60608	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	08/01/2002	**End Date:**	01/01/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Loss of muscle mass is a major health concern with prolonged space flight. Research in this field has demonstrated that, in all mammalian species studied, a loss of muscle mass (atrophy) occurs in response to decreased loading/tension associated with space flight (5). In humans, studies have suggested that decreases in rates of protein synthesis are the primary contributor to atrophy in response to unloading/disuse (5). Additionally, in cell culture, skeletal muscle atrophy during space flight results from a decrease in the rate of protein synthesis with no effect on protein degradation (25). These cell culture studies indicate that the decrease in protein synthesis is not the result of changes in circulating levels of factors such as growth hormone, glucocorticoids, and anabolic steroids but instead suggest that skeletal muscle itself can directly sense the low gravity environment. Thus, tension appears to directly regulate protein synthesis, and this is a fundamental pathway in the maintenance of skeletal muscle mass. The overall hypothesis for this study is that strain activates mechanosensitive molecules that utilize the phosphoinositide 3-kinase (PI3K) to regulate protein synthesis. This hypothesis will be addressed with the following aims.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results from this project are important for the basic understanding of how a cell regulates its size in response to changes in tension. The development of a cell culture system that produces an increase in protein synthesis in response to an increase in tension is ideal for dissecting out the signaling pathways that regulate this response. With the identification of key signaling components, we can work upstream to determine the mechanosensitive molecule(s) that initiate the signaling response. Identifying these mechanosensitive molecule(s) will be extremely valuable for the design of pharmacological treatments that can attenuate or reverse muscle atrophy
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)